Crude distillation unit (U-100) operating manual

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1. Purpose and scope

This manual covers the crude distillation unit U-100: the Feed, Heat exchange, Furnaces and Distillation areas. It describes the process, the main controls, the operating limits, start-up, shutdown and common upsets. Loop details are in the control narratives (DOC-CN-001), equipment data in DOC-DS-001 and the P&ID descriptions in DOC-PID-001.

Values in this manual are those of the plant's training simulator, which the PlantCraft demonstration reads over OPC UA.

2. Process description

The unit separates crude oil into its main cuts by atmospheric distillation. The design capacity is 12,000 m^3/d.

Crude is pumped from the crude charge tank TK-1001 by the charge pump P-1001. Three flow loops on the charge pump discharge share the charge: feed flow FIC-1001, charge flow FIC-1007 and production flow FIC-1112. The crude is preheated in the feed/product exchanger E-1009 against hot product (outlet temperature TIC-1009, product temperature TIC-1111) and then heated in the atmospheric furnace F-1104 (12 burners, fuel gas, 42 MW).

The furnace outlet temperature (TIC-1209) and the furnace temperature (TIC-1104) are controlled by the fuel gas flow FIC-1210. The heated crude enters the atmospheric column C-1101 (48 trays, 6.2 m diameter, 52 m high), controlled by the column charge flow FIC-1101. The top temperature TIC-1213 sets the cut point of the overhead product and is controlled by the reflux flow FIC-1102 from the reflux pump P-1102. A side product is drawn under FIC-1214.

The overhead vapours condense into the overhead accumulator D-1106, which holds the accumulator level LIC-1106, the reflux drum level LIC-1113 and the flash drum level LIC-1211.

3. Control philosophy

- The furnace outlet temperature is the key variable for the yield of the cuts. A change of feed rate is always made together with a change of fuel gas, never alone.
- TIC-1209 reads the thermocouple TT-1209 in the furnace outlet. That thermocouple drifts after long service at high temperature (DOC-TG-001, section 4.6). Compare it every shift with the second outlet thermocouple and with the column flash zone temperature.
- The fuel gas flow FIC-1210 has its setpoint changed often, by operating campaign. That is expected (DOC-TG-001, section 4.7).
- The reflux flow FIC-1102 and the side draw flow FIC-1214 have valves with graphite packing and known stiction (DOC-TG-001, section 4.1). A saw-tooth output on these loops is a valve problem, not a tuning problem.

- The reflux drum level LIC-1113 runs high with its outlet valve near fully open (saturation, DOC-TG-001, section 4.3).
- The flash drum level LIC-1211 uses averaging control: the level may move within its band to absorb feed variations.

4. Operating limits

Variable	Low alarm	Normal	High alarm	Action
Feed flow (FIC-1001)	40 m³/h	55 m³/h	70 m³/h	low-low 30 m³/h: furnace low-flow trip
Furnace outlet temperature (TIC-1209)	62 °C	69 °C	75 °C	80 °C: fuel gas trip
Furnace temperature (TIC-1104)	60 °C	68 °C	74 °C	
Fuel gas flow (FIC-1210)	30 Nm³/h	57 Nm³/h	80 Nm³/h	
Column top temperature (TIC-1213)	55 °C	60 °C	65 °C	
Reflux flow (FIC-1102)	35 m³/h	48 m³/h	60 m³/h	
Overhead accumulator level (LIC-1106)	20 %	52 %	80 %	
Reflux drum level (LIC-1113)	30 %	84 %	92 %	

The furnace low-flow trip protects the tubes: a furnace pass without flow overheats in minutes.

5. Start-up (summary)

1. Circulate cold crude through the furnace and column bottom with FIC-1001 at 40 m³/h.
2. Light the furnace pilots, then the burners one by one. Raise the outlet temperature at no more than 50 °C per hour.
3. Establish the reflux with P-1102 when the accumulator D-1106 reaches 30 % level.
4. Put TIC-1213 in automatic when the top temperature is stable.
5. Raise the charge to the normal rate in steps of 5 m³/h.

6. Normal shutdown (summary)

Reduce the charge to 40 m³/h, reduce the furnace outlet temperature at no more than 50 °C per hour, put out the burners, and keep circulation until the furnace outlet is below 40 °C.

7. Common upsets

7.1 Furnace outlet temperature does not match the column

If the furnace outlet temperature TIC-1209 reads at setpoint but the column flash zone is colder than usual, suspect drift of TT-1209 (calibration interval 6 months, DOC-DS-001). Do not chase the reading with more fuel. Ask instrument maintenance for a check against a portable thermocouple.

7.2 Exchanger outlet temperature swings

TIC-1009 swings after changes of crude tank or of the product rate through E-1009 (load disturbances, DOC-TG-001 section 4.8). The swings settle in 10 to 15 minutes. Exchanger fouling increases the swings over the months between cleanings.

7.3 Reflux flow cycling

FIC-1102 cycles with a saw-tooth output. This is the stiction of FV-1102. The column top temperature TIC-1213 follows the cycle. Ask for a valve travel test.